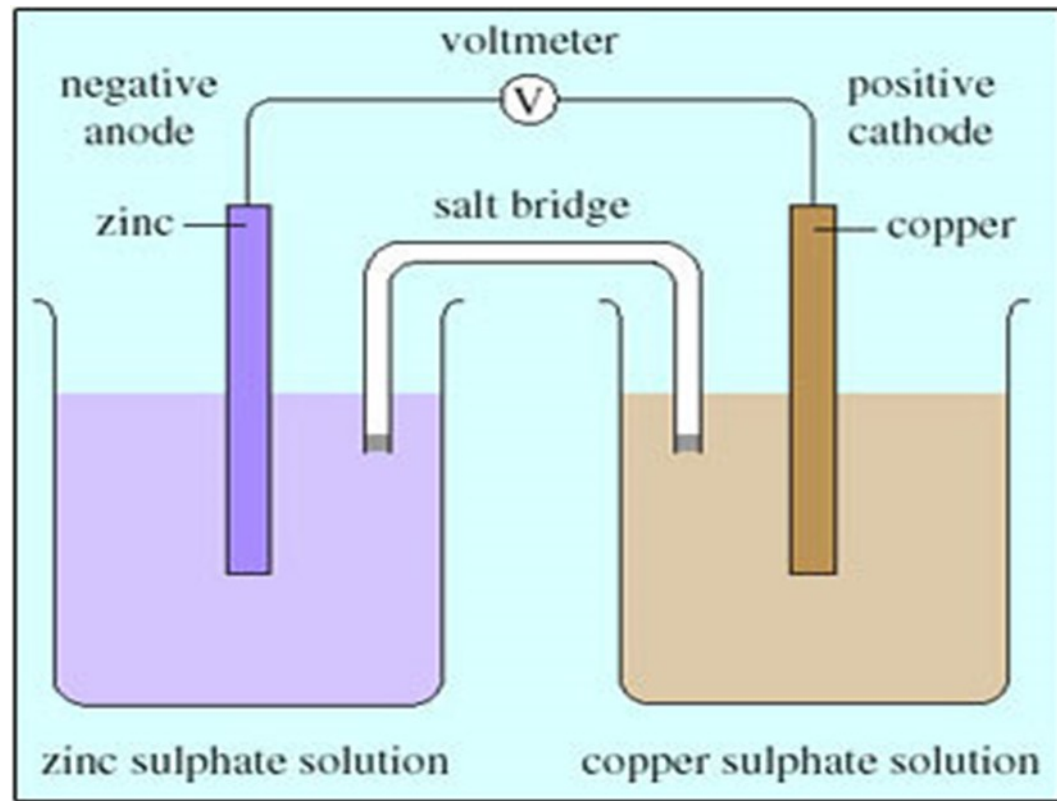


ELECTROANALYTICAL METHODS



Outline

- ☐ INTRODUCTION
- ☐ POTENTIOMETRY
- ☐ POLAROGRAPHY
- ☐ CONDUCTOMETRY
- ☐ COULOMETRY

INTRODUCTION

- **Electroanalytical techniques** are concerned with the measurement of **current, potential or charge** and their relationship to **chemical parameters** such as **concentration**.

✓ **With the help of instruments potentiometer, conductometer, voltameter**

- **Potential,**
 - **Current, or**
 - **Charge**
- 
- Serves as the analytical signal**

INTRODUCTION...

Classification of Electroanalytical Techniques

- ✓ **Bulk methods**- due to property of the whole solution.
 - Example: measurement of a solution's conductivity, which is proportional to the total concentration of dissolved ions.
- ✓ **Interfacial methods**- the signal is a function of phenomena occurring at the interface between an **electrode** and the **solution** in contact with the electrode.

Interfacial Electrochemical methods

- **Static Methods:**

- **No current passes in the electrochemical cell**

- e.g. Potentiometry**

- **Dynamic Methods:**

- **Current flows in the electrochemical cell**

- Constant Current Methods**

- ✓ **Fixed current allows complete oxidation or reduction of the analyte. Quantity of electricity (**Coulometry**) or material deposited on electrodes (**Electrogravimetry**) are determined**

Interfacial Electrochemical methods....

□ Controlled-Potential Methods:

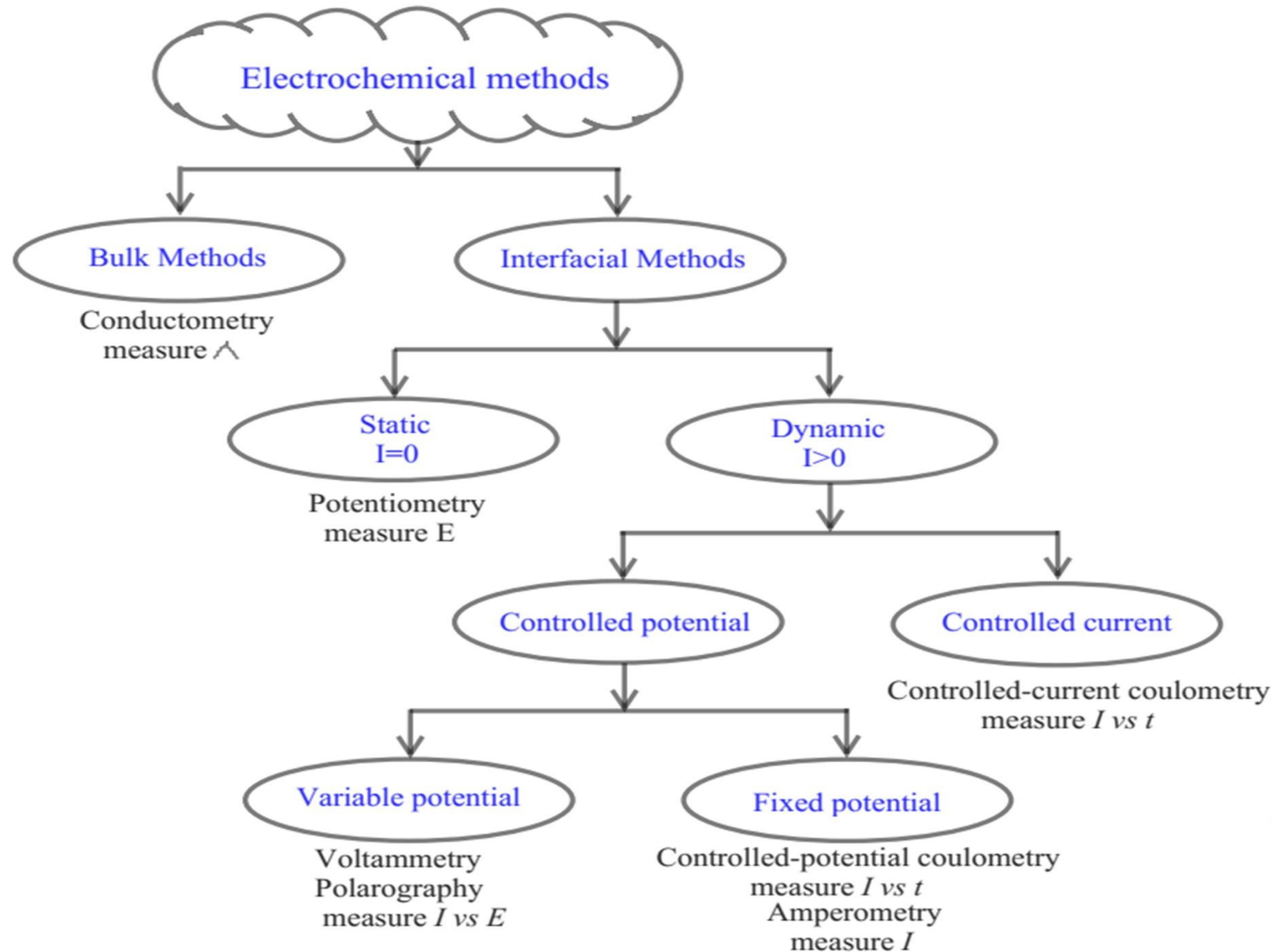
✓ Constant potential is applied during analysis while measurements of other variables are carried out.

Sensitive and can be carried out for small volumes.

✓ For example: polarography (**Voltammetry**):

Potential is systematically varied during analysis

INTRODUCTION...



INTRODUCTION...

Terminology:

☞ ***Electrolytes***: Substances that conducts electric current as a result of a dissociation into positively and negatively charged particles called ions.

☞ ***Cations***: Positively charged ions

☞ ***Anions***: Negatively charged ions

INTRODUCTION...

☞ **Electrode:** An electrode is a solid electric conductor or that carries electrons;

☞ it can be an ***anode*** or a ***cathode***.

☞ **Anode:** *oxidation*

☞ **Cathode:** *reduction*

☞ **Electrochemical cell:** electrochemical measurements **Contains Cathode and Anode** compartments.

INTRODUCTION...

- The **cathode** is the **positive part** of the electrochemical cell as it **consumes electrons**.
- The **anode** is the **source of electrons**, making it the **negative part** of the electrochemical cell.

INTRODUCTION...

- The simplest electrochemical cell uses two **electrodes**.
 - ✓ **Indicator electrode and reference electrode**
- **Indicator electrodes**- selectively responds to changes in the **activity of the analyte being measured**
- **Reference electrodes**- its composition is fixed and **its response is stable over time. Eg standard hydrogen electrode (SHE)**

INTRODUCTION...

Terminology...

☞ **Half-cell:**

- ☞ the **anode** or the **cathode** compartment and
- ☞ **all reactions** that occur at that particular electrode.

☞ **Half-reaction:** the redox reaction that occurs in one *half-cell*

☞ **Interface (Junction):**

- ☞ the location where two distinct phases/or components come in contact with each other:

☞ **Salt Bridge**

☞ **Reduction** and **oxidation**

☞ **Oxidizing Agent** and **reducing agent**

Salt bridge and its function

- Salt bridge is usually an inverted U-tube filled with concentrated solution of electrolytes.
 - like KCl, KNO₃, and K₂SO₄ etc.

Function of salt bridge:

1. It connects the solutions of two half cells and complete the cell circuit.
2. It keeps the solutions in the two half-cells, electrically neutral.
3. It prevents liquid-liquid junction potential.

INTRODUCTION...

- The experimental system assembled all together to form electrochemical cell.

Electrochemical cells:

- are two electrodes (usually metallic) immersed in the **same electrolyte solution** or in two **different electrolyte solutions** which are in electrical contact (electrical circuit).

INTRODUCTION...

- *The arrangement of such cell is known as electrochemical cell.*
- The following figure shows an electrochemical cell for the **redox reaction**

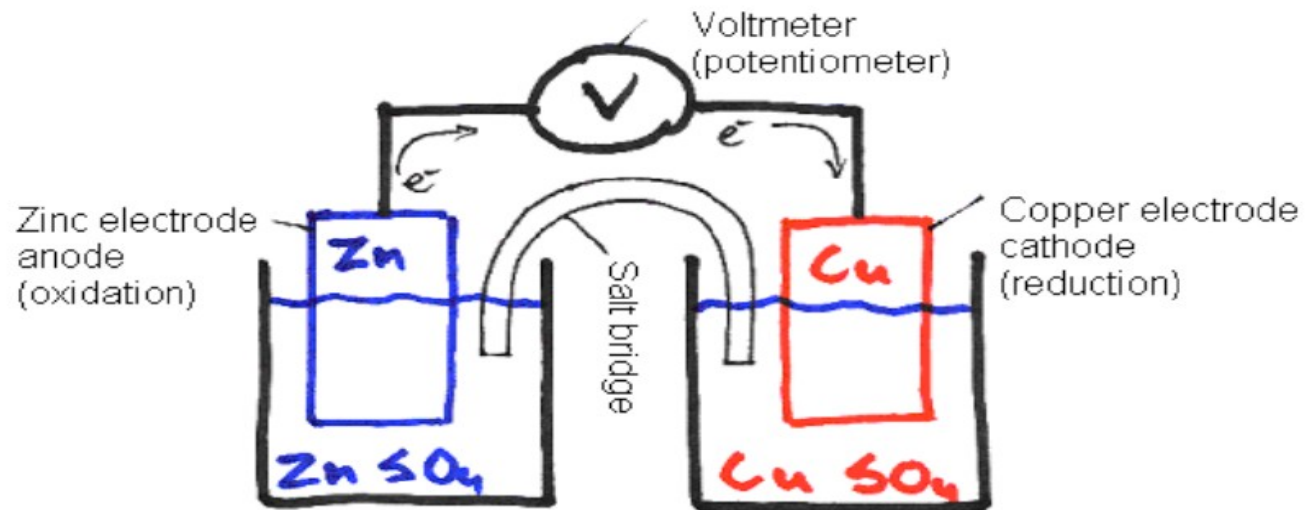
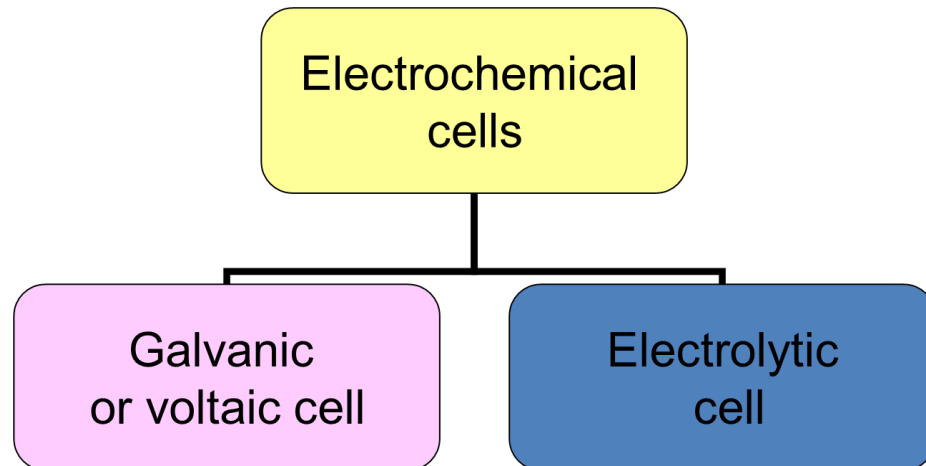


Figure 1: Electrochemical cell

INTRODUCTION...

- Conduction of electricity from one electrolyte solution to the other occurs by migration of ion in the salt bridge.
- Electrochemical cells are classed into two groups as shown in the diagram below.



INTRODUCTION...

Types of Electrochemical Cells

A. voltaic/galvanic/ and

B. electrolytic.

A. Voltaic or galvanic cells

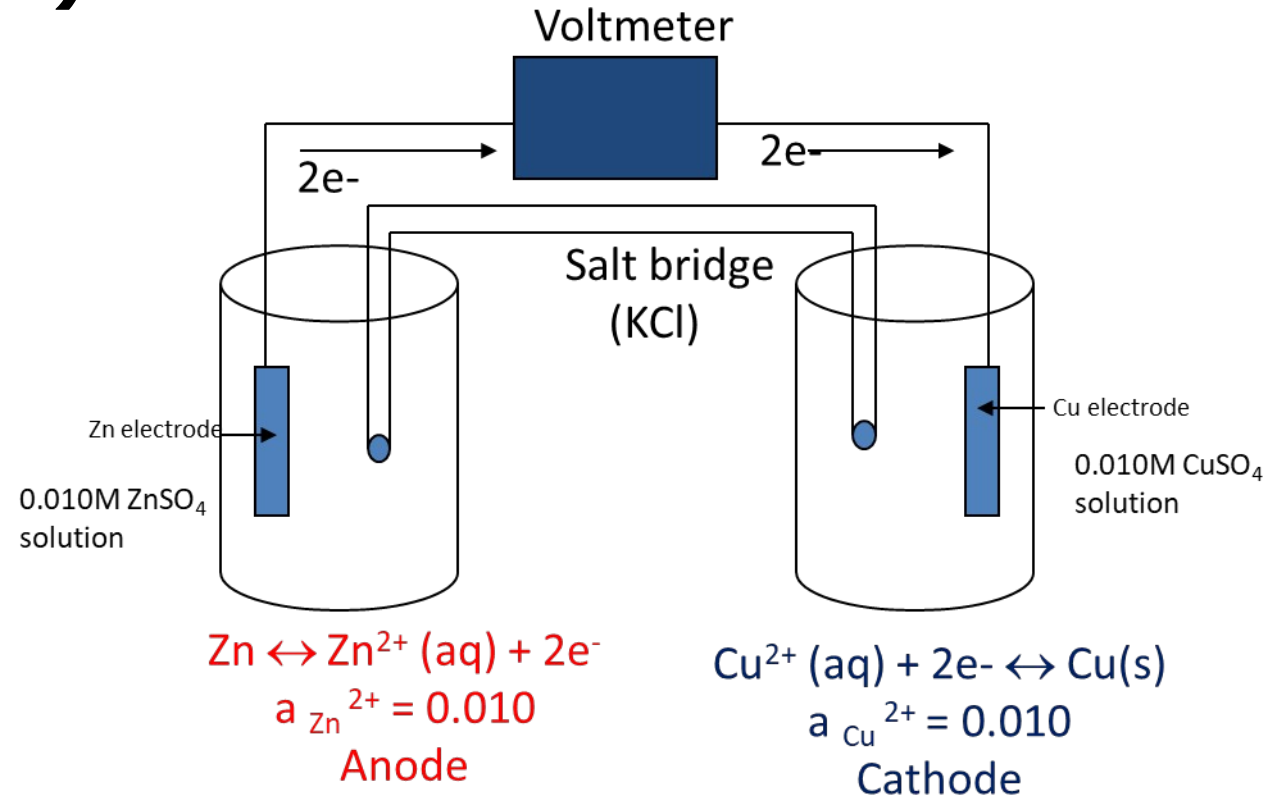
- The electrochemical reactions occur **spontaneously** when the two electrodes are connected by a conductor.
- Used to convert the **chemical energy into electrical energy.**

INTRODUCTION...

- **Metals differ in their tendency to lose electrons**
- **Zinc more readily loses electrons than copper,**
- **So, placing zinc and copper metal in solutions of their salts can cause electrons to flow through an external wire which leads from the zinc to the copper.**

INTRODUCTION...

- **Zinc (Zn) wants to ionize more than copper (Cu).**



INTRODUCTION...

- In order for the voltaic cell to **continue to produce an external electric current**, there must be a movement of ions from the **salt bridge to solution** to balance the electron flow in the external circuit.
- Recall that in order for oxidation to occur, there must be a corresponding reduction reaction that is linked or "coupled" with it.

INTRODUCTION...

- **Moreover, in an isolated oxidation or reduction half-cell, an imbalance of electrical charge would occur,**
 - **The anode would become more positive as zinc cations are produced, and**
 - **The cathode would become more negative as copper cations are removed from solution**

INTRODUCTION...

- This problem can be solved by using a **"salt bridge"** connecting the two cells as shown in the diagram above
- A "salt bridge" is a porous barrier which prevents the spontaneous mixing of the aqueous solutions in each compartment, but allows the migration of ions in both directions to maintain **electrical neutrality**

INTRODUCTION...

- The two reactions are typically written
 - $\text{Zn(s)} \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^-$
 - $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu(s)}$
- Each reaction represents a half-cell reaction.
- Schematic representation of the above cell is as follows:



The anode

The cathode

Rules which are followed in the representation of cells are:

- **Conventional chemical symbols are used to designate different chemical species such as ions, atoms or molecules**
- **Partial pressures of gases and ionic concentrations are written in parentheses following the respective symbols**

Rules which are followed....

- **Two ions placed in the same solution are separated by a comma(,)**
- **A single vertical line (/) represents a metal electrolyte boundary at which a potential difference is taken into account**
- **A double vertical line (//) denotes a salt bridge.**

INTRODUCTION...

Electrolytic cells:

- **In which chemical reactions are caused to occur by the imposition of an external voltage.**
- **Used to carry out chemical reactions at the expense of electrical energy.**

INTRODUCTION...

Similarities b/n Electrolytic cells and voltaic (galvanic) cells in the sense that both,

- ✓ Have a cathode and anode side, and
- ✓ Have a consistent flow of electrons from the anode to the cathode.

INTRODUCTION...

Differences

Electrochemical cell (Galvanic Cell)	Electrolytic cell
A Galvanic cell converts chemical energy into electrical energy.	An electrolytic cell converts electrical energy into chemical energy.
Here, the redox reaction is spontaneous and is responsible for the production of electrical energy.	The redox reaction is not spontaneous and electrical energy has to be supplied to initiate the reaction.
The two half-cells are set up in different containers, being connected through the salt bridge or porous partition.	Both the electrodes are placed in a same container in the solution of molten electrolyte.
Here the anode is negative and cathode is the positive electrode. The reaction at the anode is oxidation and that at the cathode is reduction.	Here, the anode is positive and cathode is the negative electrode. The reaction at the anode is oxidation and that at the cathode is reduction.
The electrons are supplied by the species getting oxidized. They move from anode to the cathode in the external circuit.	The external battery supplies the electrons. They enter through the cathode and come out through the anode.

ELECTRICAL CONDUCTION

- **Metallic conduction**

- Due to the flow of electrons

- **Ionic or electrolytic conduction**

- Ionic motion transports the electrons.
 - Cations move toward the negative electrode, cathode.
 - Anions move toward the positive electrode, anode